LeetCode 1295: Find Numbers with Even Number of Digits

1. Problem Title & Link

Title: LeetCode 1295: Find Numbers with Even Number of Digits

Link: https://leetcode.com/problems/find-numbers-with-even-number-of-digits/

2. Problem Statement (Short Summary)

Given an integer array nums, return the number of integers that contain an **even number of digits**.

Example

Input:

nums = [12,345,2,6,7896]

Digit counts:

12 -> 2 digits ✓

345 -> 3 digits ✗

2 -> 1 digit ✗

6 -> 1 digit ✗

7896 -> 4 digits ✓

Output:

2

Because only:

12 and 7896

have an even number of digits.

3. Examples (Input → Output)

Example 1

Input:

nums = [12,345,2,6,7896]

Output:

2

Example 2

Input:

nums = [555,901,482,1771]

Output:

1

Explanation:

555 -> 3 digits

901 -> 3 digits

482 -> 3 digits

1771 -> 4 digits ✓

Example 3

Input:

nums = [10,100,1000,10000]

Output:

2

4. Constraints

1 <= nums.length <= 500

1 <= nums[i] <= 10^5

Important:

Need count of numbers

NOT the numbers themselves.

5. Core Concept (Pattern / Topic)

Math / Digit Counting

Secondary Topics:

- Number Manipulation
- Array Traversal

6. Thought Process (Step-by-Step Explanation)

Understanding the Problem

For every number:

1. Find how many digits it contains.
2. Check if digit count is even.
3. Count it if true.

Example

12

Digits:

2

Even:

Yes

Count it.

Example

345

Digits:

3

Even?

No

Ignore.

Brute Force Idea

Convert each number to a string.

Example:

len(str(7896))

Result:

4

Even.

Count it.

Better Observation

Since constraints are small, we can simply count digits and check:

digitCount % 2 == 0

Key Insight

The problem is just:

Count numbers whose digit length is even.

7. Visual / Intuition Diagram (ASCII Diagram)

```
Input:
[12,345,2,6,7896]
Process:
12   -> 2 digits -> ✓

345  -> 3 digits -> ✗

2    -> 1 digit  -> ✗

6    -> 1 digit  -> ✗

7896 -> 4 digits -> ✓
```

```
Count:
2
```

8. Pseudocode (Language Independent)

```
count = 0

for each num

    digits = number of digits

    if digits is even

        count++

return count
```

9. Code Implementation

Python

Method 1: String Conversion (Simple)

```
class Solution:
    def findNumbers(self, nums):

        count = 0

        for num in nums:

            if len(str(num)) % 2 == 0:
                count += 1

        return count
```

Java

```
class Solution {

    public int findNumbers(int[] nums) {

        int count = 0;

        for(int num : nums) {

            if(String.valueOf(num)
                      .length() % 2 == 0) {

                count++;
            }
```

```
        }

        return count;
    }
}
```

Mathematical Solution (Without String)

Python

```python
class Solution:
    def findNumbers(self, nums):

        count = 0

        for num in nums:

            digits = 0

            while num > 0:

                digits += 1
                num //= 10

            if digits % 2 == 0:
                count += 1

        return count
```

10. Time & Space Complexity

Time Complexity

O(n)

Visit every number once.

Space Complexity

O(1)

Only variables used.

11. Common Mistakes / Edge Cases

Mistake 1

Checking:

num % 2 == 0

Problem asks:

Even number of digits

NOT:

Even value

Mistake 2

Returning the numbers themselves.

Need:

Count

only.

Mistake 3

Confusing:

100

with:

2 digits

Actually:

100 → 3 digits

Edge Cases

Single Digit

Input:

[7]

Output:

0

Two Digits

Input:

[10]

Output:

1

Maximum Value

Input:

[100000]

Digits:

6

Output:

1

12. Detailed Dry Run

Input:

nums = [12,345,2,6,7896]

Number	Digits	Even?	Count
12	2	Yes	1
345	3	No	1
2	1	No	1
6	1	No	1
7896	4	Yes	2

Return:

2

13. Common Use Cases (Real-Life / Interview)

Data Validation

Count IDs with specific length.

Number Analysis

Classify numbers by digit count.

Formatting Systems

Detect short vs long numeric codes.

Interview Purpose

Tests:

Digit Manipulation

Basic Math

Array Traversal

14. Common Traps (Important!)

Trap 1

Checking value parity instead of digit parity.

Trap 2

Overcomplicating with logarithms.

Trap 3

Forgetting that:

10000

has:

5 digits

not 4.

15. Builds To (Related LeetCode Problems)

LC 1295 - Find Numbers with Even Number of Digits

↓

LC 258 - Add Digits

↓

LC 9 - Palindrome Number

↓

LC 66 - Plus One

↓

LC 7 - Reverse Integer

These strengthen digit manipulation skills.

16. Alternate Approaches + Comparison

Approach	Time	Space	Interview Rating
String Conversion	O(n)	O(1)	Best
Mathematical Counting	O(n log M)	O(1)	Good
Logarithm	O(n)	O(1)	Advanced

Approach 1: String Conversion (Chosen)

len(str(num))

Simplest and most readable.

Approach 2: Mathematical Digit Counting

Repeatedly divide by:

10

until:

0

Approach 3: Logarithm

digits = int(log10(num)) + 1

Works but less intuitive.

17. Why This Solution Works (Short Intuition)

A number has an even digit count if the length of its decimal representation is even.

By checking every number and counting those with even digit lengths, we directly obtain the answer.

18. Variations / Follow-Up Questions

Follow-Up 1

Return the numbers themselves instead of the count.

Follow-Up 2

Count numbers with odd digits.

Follow-Up 3

Find the number having the maximum digit count.

Follow-Up 4

Group numbers by digit length.

Interview Takeaway

The key pattern is:

Digit Counting

Whenever you hear:

Number of Digits

Length of Number

Digit Properties

think:

String Conversion

or

Repeated Division by 10

This is a foundational number-manipulation pattern that appears frequently in easy interview questions.